الجامعة المصرية
للتعلم الإلكتروني الأهلية

EELU
THE EGYPTIAN E-LEARNING UNIVERSITY

Advanced Computer Networks Section 2

What is UDP?

User Datagram Protocol – A connectionless transport layer protocol

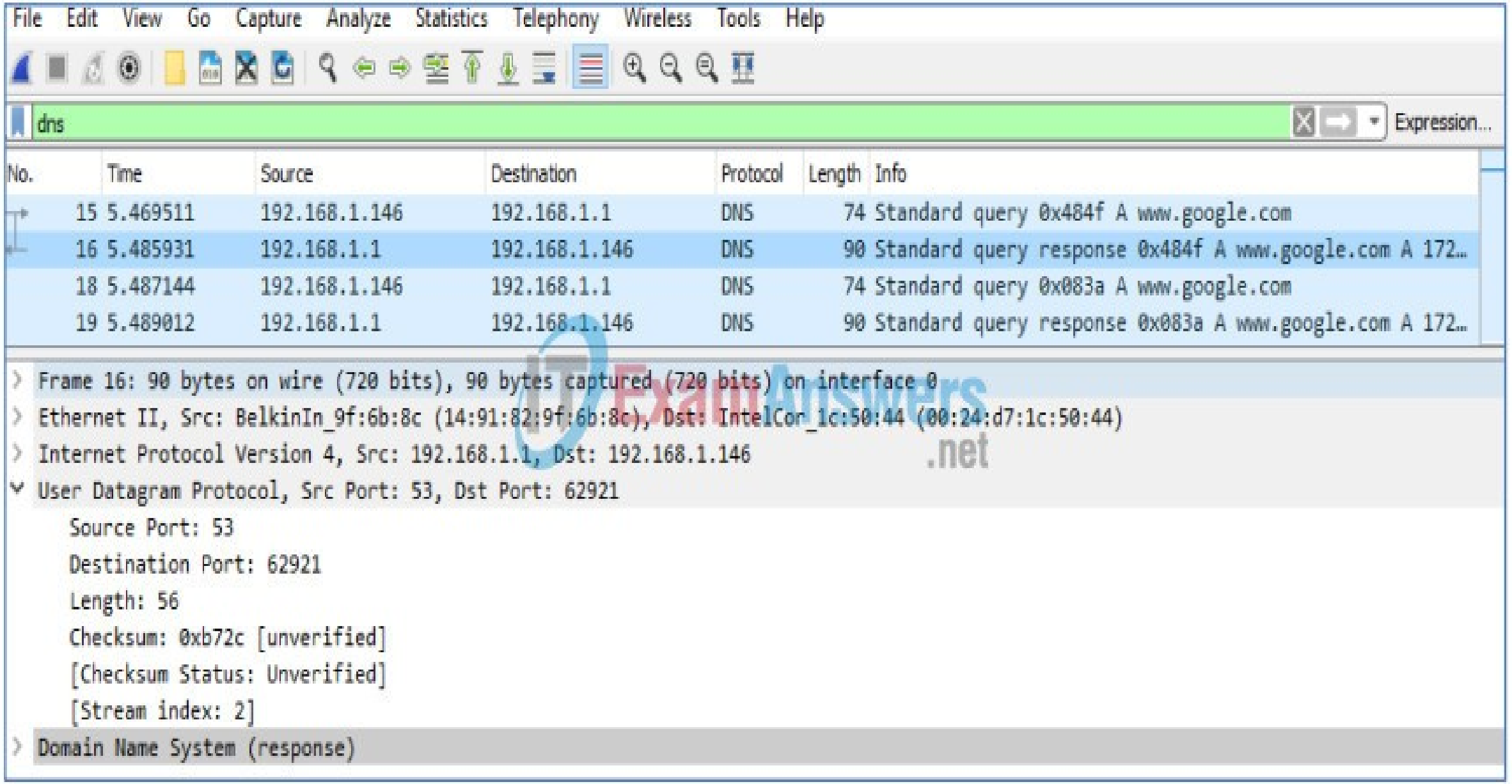
Features:

- No handshaking
- No guarantee of delivery
- No congestion control
- Low overhead, faster than TCP

Common Uses: DNS

UDP Header Structure

Only 4 fields, 8 bytes total.
Source Port & Destination Port (2 bytes each): Identify the sending and receiving application.
Length (2 bytes): The total length of the UDP datagram (header + data).
Checksum (2 bytes): Optional for IPv4, mandatory for IPv6. Used for error-checking the header and data.


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
Expression...
No. Time Source Destination Protocol Length Info
15 5.469511 192.168.1.146 192.168.1.1 DNS 74 Standard query 0x484f A www.google.com
16 5.485931 192.168.1.1 192.168.1.146 DNS 90 Standard query response 0x484f A www.google.com A 172…
18 5.487144 192.168.1.146 192.168.1.1 DNS 74 Standard query 0x083a A www.google.com
19 5.489012 192.168.1.1 192.168.1.146 DNS 90 Standard query response 0x083a A www.google.com A 172…
Frame 16: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface 0
Ethernet II, Src: BelkinIn_9f:6b:8c (14:91:82:9f:6b:8c), Dst: IntelCor_1c:50:44 (00:24:d7:1c:50:44)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.146
User Datagram Protocol, Src Port: 53, Dst Port: 62921
Source Port: 53
Destination Port: 62921
Length: 56
Checksum: 0xb72c [unverified]
[Checksum Status: Unverified]
[Stream index: 2]
Domain Name System (response)

Connectionless Nature (No "Conversation")

No Handshake: Unlike TCP, there is no SYN, SYN-ACK, ACK to establish a connection.
No Acknowledgment: Packets are sent without expecting an ACK.
No Retransmission: If a packet is lost, UDP does not try to send it again.

How to Capture UDP Traffic in Wireshark

Open Wireshark → Select Interface
Start Capture
Generate UDP Traffic (e.g., nslookup example.com)
Stop Capture
Use Filter: **udp**

Questions

From your trace, what is the size of the UDP header?

Does UDP include any error recovery mechanism?

Which application uses UDP port 53?

Compare the overhead of UDP vs. TCP using your trace.

What is the maximum UDP payload size?

Using Wireshark, capture a UDP communication between two devices (for example, using DNS or a simple UDP chat application).

a) Identify the source and destination port numbers in the initial UDP request.

b) Identify the source and destination port numbers in the UDP reply.

c) Explain how the port numbers are used differently in the request and the reply.